

1. Solve the following :

- (a) $|x + 3| = |2x + 1|$,
 (b) $|x + 3| \leq |2x + 1|$.

2. Consider the line $l(x) = 5x + 3$. Prove that, for every $\epsilon > 0$, if the distance from x to 1 is smaller than $\frac{\epsilon}{5}$; then, the distance from $l(x)$ to 8 is less than ϵ .

3. Find all the possible values of $k \in \mathbb{R}$, such that the perpendicular bisector of the line segment joining the points $(1, 4)$ and $(k, 3)$ is parallel to the line $y = 5x - 20$.

4. Solve the following:

- (a) $|x| + |y| = 1$,
 (b) $|x| + |y| \leq 1$

Appendix C and D

5. Sketch the following regions in the xy -plane:

(a)

$$\{(x, y) : x, y \in \mathbb{R}, x^2 - 4x < -y^2\}$$

(b)

$$\{(x, y) : x, y \in \mathbb{R}, x^2 + 4y^2 \geq 4\}$$

(c)

$$\{(x, y) : x, y \in \mathbb{R}, x^2 - 4x < -y^2, x^2 + 4y^2 \geq 4\}$$

6. Solve the following equation:

$$2 \cos x + \sin 2x = 0$$

Sections 1.1 and 1, 2

7. Find the domain and Image of the following function?. Is it Symmetric with respect to the x -axis or with respect to the origin?

$$G(t) = \frac{2}{t^2 - 16}$$

8. Is the graph of $|x| + |y| = 1$ the graph of a function?

9. Sketch the graph of

$$f(x) = \begin{cases} \sqrt{1+x^2} & \text{if } x \geq 0 \\ -(x-1)^2 + \sqrt{2} & \text{if } x < 0 \end{cases} \quad (1)$$

Is it Symmetric with respect to the x -axis or with respect to the origin?

Section 1.3-1.5

10. Sketch the graph of $f(x) = -\cos(-x + \frac{2\pi}{3}) + 4$

11. Is the following function one to one (injective)?. If so, find its inverse:

$$f(x) = \frac{4x - 1}{2x + 3}$$

12. A bacteria culture starts with 500 bacteria and doubles in size every half hour.

- (a) How many bacteria are there after 3 hours?.
 (b) How many bacteria are there after t hours?
 (c) How many bacteria are there after 40 minutes?